

Real Time Accident Alert System in Hilly Region using LoRa Technology

Eshwari M

Department of Electronics and
Communication Engineering,
Vels Institute of Science, Technology
& Advanced Studies
Roginipapa3030@gmail.com
(VISTAS), Chennai – 600117, India

Mohana Sundaram S

Department of Electronics and
Communication Engineering,
Vels Institute of Science, Technology
& Advanced Studies
mohanasundarm2000@gmail.com
(VISTAS), Chennai – 600117, India

Guide: Dr. Arul Stephen C

Department of Electronics and
Communication Engineering,
Vels Institute of Science, Technology
& Advanced Studies
arul.se@vistas.ac.in
(VISTAS), Chennai – 600117, India

Abstract — In hilly regions, ensuring road safety is particularly challenging due to the unique terrain and the potential for more severe consequences in the event of accidents. This study aims to improve road safety in hilly regions by utilizing three distinct accident detection parameters. The first parameter, tilt sensors (MPU6050), monitors vehicle inclination, triggering an alert system when sudden changes in the tilt angle indicate a rollover or collision. The second parameter, vibration sensors, identifies abrupt jolts or impacts, triggering the alert system when unusual vibrations occur. The third parameter, GPS technology, provides accurate positioning and location data for notifying nearby helping stations and first responders, facilitating rapid and precise responses in emergency situations. To ensure efficient communication of accident alerts over long distances, LoRa (Long-Range Radio) technology is integrated into the system. This communication enables swift transmission of accident alerts to the nearest helping station. The central interface for accident detection and alerting is the Blynk mobile app, accessible to a wide range of users, including vehicle operators and first responders. This study presents a significant advancement in enhancing road safety and emergency response in hilly regions, where accidents can have serious consequences. Additionally, we plan to incorporate sending an emergency alert message to the receiver's number, nearby hospital, and police station using a GSM module to further enhance emergency response.

Keywords — Hilly region, Accident detection, LoRa (Long Range), GPS (Global Positioning System), GSM (Global System for Mobile Communication), thinks board app

I. INTRODUCTION

In the year 2022, road accidents claimed the lives of over 1.68 lakh individuals in India. To put it in perspective, on average, India witnessed 53 accidents and 19 deaths every hour, which amounts to about 1,264 accidents and 42 deaths every single day due to road crashes. Regrettably, these statistics have remained distressingly consistent in the past few years. Considering hilly regions, the average annual death toll in Himachal Pradesh over the last decade was 1,100, and in Jammu & Kashmir, it stood at 900[1-2]. Still, these accidents inflict a significant loss on individuals, society, and the entire country. The main reason for such tragedy can be classified as three components: Late accident detection, bad cellular network and no accurate location for the accident. The consequences of an accident in such hilly areas can be particularly severe, primarily due to delayed emergency responses. This delay is often caused by the inherent challenges of detecting accidents and tracking their exact locations in real-time. In addition, the

absence of a reliable network infrastructure in these areas are cause for these problems.

This research seeks to provide a solution to this critical issue. The proposed solution takes an innovative approach to leverage the possibilities of Long Range (LoRa) technology. This makes it an optimal choice for the proposed system because it can facilitate long range communication while maintaining low power usage. Depending on the transmission conditions and obstacles present, the communication range can vary from 1 to 15 km. LoRa can do high range transmission with consuming low power. The data rate transfer is from 0.3 kb/s to 27 kb/s, which depends on the factor of spreading. Also, LoRa uses adaptive power. This adaptation is dependent on parameters data rate & link conditions across others. In this proposed system, an MPU6050 6-axis Gyro/Accelerometer Sensor, NEO-6M GPS module, and SW-420 vibration sensor, all connected to the Arduino Uno, are used which acts as the transmitter system and is installed on the vehicle that needs to be tracked. Additionally, a LoRa-based transmitter is integrated into this system. The receiver system comprises a LoRa module and a NodeMCU board connected to the Blynk mobile application through Wi-Fi. The application displays live location and accident alert messages. This system is engineered to transmit real-time location data to the receiver system and automatically detect accidents using the above-mentioned sensors. Once an accident is detected, the location data is sent to the nearby hospital as well as police station from receiver system via a GSM sensor.

The primary objective of this proposed system is to provide real-time tracking of a vehicle and fasten the emergency response process by offering precise location data immediately after an accident occurs, especially in areas where cellular networks are unreliable or unavailable. In the coming sections, the design considerations, implementation details, and performance evaluation of the proposed system are explored. Through this study, the main aim is to contribute to the ongoing efforts aimed at improving road safety and reducing the risks associated with road accidents in hilly regions.

II. RELATED WORKS

G.M. Bianco, et al. [4] outlines a LoRa-enabled system for Search and Rescue (SaR) missions in difficult mountain

terrain. The integrated system consists of a Body Area Network and experimental propagation models that allow precise position pinpointing within meters and provide a specialized and efficient solution for mountainous SaR operations.

X. Zhang, et al. [5] discusses a new transmission tower tilt monitoring system that uses LPWAN technology such as LoRa modulation and NB-IoT. To overcome issues in complex surroundings, the solution employs a LoRa multi-node network to provide real-time tower tilt parameters.

M.J. Faber, et al. [6] investigates the theoretical and practical aspects of LoRa technology, examining its performance in low signal-to-noise ratio environments. The study employs a model and experiments to evaluate LoRa's communication capability, creating effective connections over three and ten kilometers in open and urban environments, respectively.

N.V.K Ramesh, et al. [7] offers a device that promptly notifies the driver's emergency contact, estimates response time, and transmits precise accident locations to the nearest hospital and police station. A buzzer alarm alerts the driver, allowing for prompt medical and law enforcement assistance and improving overall road safety.

N. Parveen, et al. [8] describes an accident detection system based on an accelerometer monitored by Arduino. If an accident occurs, Arduino communicates with a GSM module, which delivers important location information to emergency personnel via text messages. The system provides timely alarms, which are crucial in areas with limited access to emergency aid, as well as insights into the dynamics of the event.

S. Gatade, et al. [9] describes a quick car accident detection system based on the ESP32 microcontroller, which includes Wi-Fi and Bluetooth. Sensors that detect sudden stops, passengers, objects, and suspected driver impairment use infrared, ultrasonic, and accelerometer technology. The technology, which is GPS and GSM enabled, delivers notifications and accurate accident locations to healthcare providers or specified contacts via the Blynk mobile app, thereby improving reaction times and overall safety monitoring.

Bhandari, et al. [10] proposes an Internet of Things (IoT) system to solve safety problems in hilly tourist areas. Recognizing the increasing number of tourism-related accidents and landslides, the study advocates for technological interventions. The Internet of Things-based method focuses on efficient and cost-effective accident detection and location identification, potentially saving lives for both inhabitants and visitors in these locations.

A.H. Alkhawani, et al. [11] proposes an IoT-based intelligent security architecture for rapid and secure accident detection and reporting, with a focus on driver privacy. The system employs design science research to

enable speedy accident detection, data encryption, and GSM transfer. The study investigates the security and privacy hazards associated with Internet of Things devices, providing critical information for Saudi Arabia's accident response systems and emphasizing the potential to save lives and enhance road safety.

R, et al. [12] proposes addressing road safety concerns with Wireless Sensor Networks (WSN) and the Internet of Things (IoT), particularly in mountainous terrain. The technique explains the algorithm for prioritizing vehicle movement on bends, with potential real-time updates via an Android app, adding to better safety measures in poor road conditions.

K. Nandanija, et al [13], uses long-range (LoRa) technology to be applied for emergency response and real-time accident detection, particularly in isolated and mountainous locations. This approach relays data to a receiver system connected to a mobile app by utilizing LoRa technology in conjunction with automobile sensors. LoRa's longer range and low power consumption allow for reliable communication in difficult environments, improving traffic safety in hilly places.

K.V, et al [14], shows the growth of IoT devices worldwide is driving up demand for wireless sensor networks. With LoRa technology and a Beat Sensor at its core, it is powered only by ambient energy harvesting and reaches an amazing 5.25-kilometer communication range with low temperature inaccuracy.

S. Chandran, et al. [15] examines the application of Long Range (LoRa) technologies in smart cities. LoRa's long range and low power consumption are touted as benefits for a wide range of IoT applications. LoRa technology, which works at sub-GHz frequencies, reduces interference while maintaining signal transmission consistency and energy economy. The research looks at the efficacy of LoRa in Kuala Lumpur's mountainous suburbs, taking altitude and spreading effects into account.

P. Veeraragaavan, et al. [16] provides a unique solution to the traffic delays suffered by emergency vehicles. When an accident is detected, the system dispatches a licensed ambulance to the nearest hospital that provides emergency care. This efficient ambulance system, which employs GPS, collision sensors, and LoRa technology, aims to deliver speedy and effective emergency response, potentially saving many lives.

P. Sarma, et al. [17] describes an all-encompassing accident detection and prevention system implemented in Python. The system includes accelerometer/gyro sensors, ultrasonic and vibration sensors, as well as Arduino, GPS, and GPRS modules. The ultrasonic sensors identify obstructions, the vibration sensor detects vehicle movement, and the collected data is used to trigger GSM module messages via Arduino.

J. Thomas, et al. [18] focuses on LoRaWAN version 1.1 security concerns, namely the physical layer's vulnerability to Man-in-the-Middle (MitM) attacks during communication between two modules. The paper describes how such attacks occur and suggests a solution to this security problem in LoRaWAN communication.

D.H. Kumar, et al. [19] seeks to use GSM technology to immediately identify accidents and transmit the location of the incident. The accident signal identification system employs a number of sensors, including alcohol, vibration, and fire detectors. After detecting something, the microcontroller sends an SMS to pre-programmed contacts with the vehicle's coordinates and activates safety features such as airbag deployment and brake locking. Because of this guarantee of prompt assistance, aid and ambulance services can arrive at the accident scene swiftly.

M.L.S. Kumar, et al. [20], a simulation model for using sensors to identify accidents is introduced. The Proteus Simulation Tool displays the circuit simulation. Upon discovering the disaster, friends and family receive a message. The Android application prototype, which aids in accident detection, is created using the Blynk platform.

Visnja Krizanovic, et al. [20], investigate the implementation of an energy-efficient environmental monitoring system for precision agriculture using LoRa-based wireless sensor networks (WSNs). The system utilizes LoRa technology for long-range, low-power communication across agricultural fields. The study demonstrates significant energy consumption reduction and reliable data transmission, showcasing LoRa's potential in enhancing sustainability and efficiency in agricultural monitoring.

III. METHODOLOGY

This system makes an extensive use of lora technology. Figure 1 explains the main architecture for the proposed system. Firstly, the main controller, an Arduino Uno, is used to establish connection with the sensors. To record real-time data, sensors such as the SW-420 Vibration Sensor, MPU6050 Gyro-Accelerometer, and NEO-6M GPS Module were connected to the Arduino Uno. The LoRa module enabled wireless connectivity, allowing for long-distance data transmission in tough terrain. Furthermore, the NodeMCU ESP8266 was configured as a receiver to connect to the Blynk app via Wi-Fi for data display purpose. In the second phase each sensor is integrated together accordingly to provide precise data points needed for accident detection. The SW-420 Vibration Sensor, which is used to detect vibrations, played an important role in detecting vibrations if accident has happened. The MPU6050 Gyro-Accelerometer was used to capture changes in orientation and movement, allowing for a thorough understanding of the vehicle's dynamics. Also, this sensor gives value of surrounding temperature of environment. This value will be useful in case if fire caught by car. These 2 sensors help to detect accident of the vehicle. The NEO-6M GPS Module enabled in exact location tracking, permitting calculation of the accident's geographical coordinates. These sensors collaborated together to build a powerful network, monitoring and passing data to the central unit on a continuous basis. The third phase focused on creating an efficient communication between transmitter and receiver. Because of long-range capabilities and low power consumption, LoRa technology was chosen for data transfer between the transmitter and receiver. This was particularly significant in hilly locations with rough terrain and few connectivity options. The seamless integration of LoRa modules ensured consistent and real-time data delivery, overcoming geographic barriers. The Wi-Fi-enabled NodeMCU ESP8266 expands the system by connecting to the Blynk app, allowing users to access crash data and call notifications in real time.

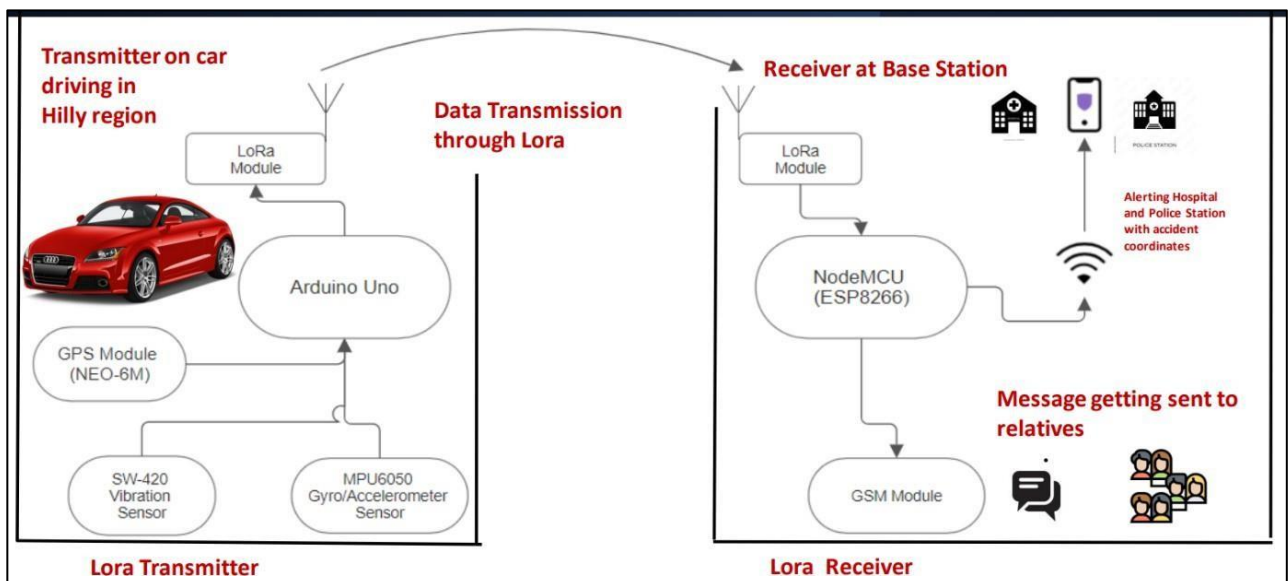


Fig. 1. Proposed Accident Alert System

The final phase involves testing the system functionality. A variety of scenarios, such as possible collapse scenarios, are considered to evaluate the system behavior and accuracy. An integrated GSM Module has been tested for SMS alerts to ensure timely notification. Response time, data quality, and stability of the interaction were all carefully assessed. Iterative testing and user feedback played a key role in designing the system and verifying its effectiveness in providing real-time accident warnings in challenging hilly terrain. The threshold defined for this system are for vibration, temperature and angle. If angle of the car is more than 20 ° or vibration is more than 1000Hz or temperature of the car is more than 100° C (in case of fire) then it indicates that an accident is happened.

The main architecture behind the proposed system is represented in the figure 2. For this system, the workflows follow firstly the accident detection. The accident is detected by the sensors presented and the GPS location is collected by GPS sensor. In this GPS technology, location is determined by using a network of satellites that orbit the Earth. A GPS receiver calculates its position by precisely timing the signals sent by these satellites. So these both (accident & location) data is then transmitted to the receiver LoRa. The data is then evaluated and the message, if accident happens, is communicated to the nearby helping stations.

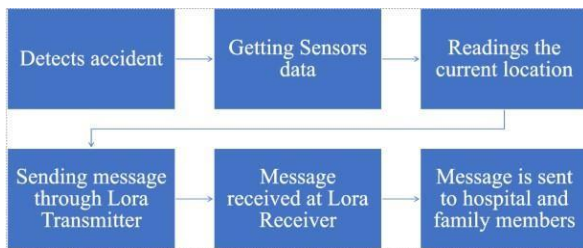


Fig. 2. Architecture: Event-Driven Architecture

IV. IMPLEMENTATION

The below Table I shows the technical specifications of hardware used in this system.

TABLE I. HARDWARE SPECIFICATIONS

Hardware Components	Specifications
1.Arduino Uno (Main Controllor)	Operating Voltage: 5V Input Voltage: 7-12V Digital I/O Pins: 14 (of which 6 provide PWM output) USB Interface: ATmega16U2
2. SW-420 Vibration Sensor	Operating Voltage: 3.3V - 5V Current Consumption: <1mA Sensitivity Adjustment: Potentiometer for sensitivity adjustment Detection Range: Adjustable (depends on sensitivity setting)

3.MPU6050 Gyro-Accelerometer	Operating Voltage: 3.3V - 5V Gyroscope Range: ± 250 , ± 500 , ± 1000 , or ± 2000 degrees per second Accelerometer Range: $\pm 2g$, $\pm 4g$, $\pm 8g$, or $\pm 16g$ Digital Output: I2C (Inter-Integrated Circuit) communication protocol
4. NEO-6M GPS Module:	Receiver Type: 50-channel u-blox 6 positioning engine Operating Voltage: 3.3V - 5V Current Consumption: Approximately 20mA Positioning Accuracy: Approximately 2.5 meters (C/A code, without aiding) Update Rate: Up to 5Hz (configurable)
5. LoRa Module	Operating Frequency - 433 MHz Sensitivity - 148 dBm. Data rates range from 0.3 kbps to 37.5 kbps A SPI interface for communication with other devices. Operating voltage - 1.8V to 3.7V
6. NodeMCU ESP8266:	Microcontroller: Based on the ESP8266EX microcontroller, which is a low-cost Wi-Fi chip GPIO Pins: Typically, around 10 to 11 pins Wi-Fi Connectivity: Supports 802.11 b/g/n Wi-Fi standards Operating Voltage: 3.3V
7. GSM 900A Module	Supports dual-band frequencies, typically 890/915/1800/25 MHz UART (Universal Asynchronous Receiver-Transmitter) interface for communication with microcontrollers or other devices Operating voltage: 5V.

Arduino is chosen here because of its ease of use and it is being supported by wide online community that provides extensive resources, tutorials, and troubleshooting help. The transmitter node, also known as the accident node, is put on a vehicle traveling hilly terrains. Figure 3 explains the transmitter node connections. LoRa is needed for this system for its ability to transmit data over long distances with low power consumption, making it ideal for the rural or remote areas where cellular networks may be unreliable or unavailable. The system has three integrated sensors: the

SW-420 Vibration Sensor for detecting sudden vibrations, the MPU6050 Gyro-Accelerometer for tracking changes in orientation and movement, and the NEO-6M GPS Module providing exact location coordinates. These sensors are linked to the Arduino Uno, which serves as the central processing unit. The acquired data is then wirelessly sent using the LoRa Module, which is known for its long-range capabilities, ensuring reliable connectivity even in tough topographies. As the vehicle travels, the LoRa transmitter module continuously sends real-time data to the base node for analysis and accident detection.

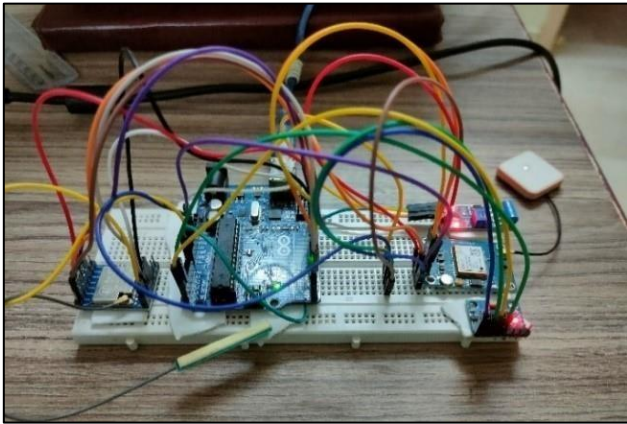


Fig. 3. Transmitter Node

The receiver system at the base node is responsible of the receiver system at the base node responsible of evaluating the transmitted data. Figure 4 explains the receiver node connection. The base node is a robust accident detection mechanism comprised of a LoRa Module for communication reception, a NodeMCU ESP8266 for data analysis, and a GSM GA6 Module for alert generating. When the LoRa Module receives data from the transmitter node, it sends it to the NodeMCU ESP8266, which uses established algorithms to detect patterns suggestive of an accident. When an accident is detected, the GSM Module initiates an alarm system by sending SMS alerts and call to designated contacts. In addition, an Arduino Uno is used to power the GSM module, assuring uninterrupted performance.

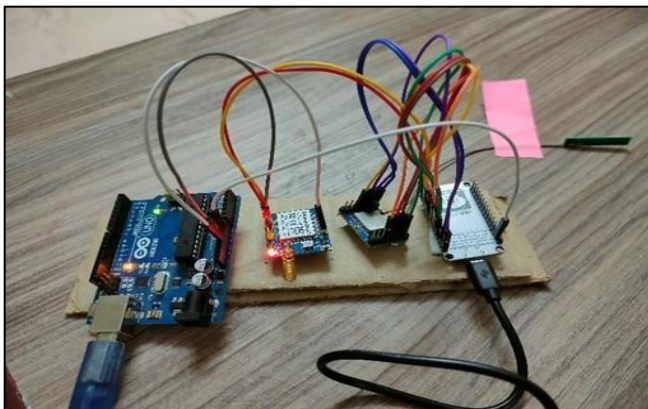


Fig. 4. Receiver Node

The integration of the Things board app improves the user experience by giving receiver side user a visual depiction of the system's status and encouraging involvement. Figure 5 explains the Blynk app configuration. User can track the vehicle's real-time data, such as vibration levels, orientation changes, and exact GPS coordinates, via the Blynk app. The software offers an easy-to-use interface for tracking the vehicle's travel in the event of an accident. But the app becomes an indispensable tool for quick action when an accident is recognized. Receiver side user of the Blynk app will get notified of the emergency through visual notifications triggered by the accident alarm. This user- friendly interface is a crucial element that improves the entire experience of the user and guarantees effective communication with the real-time accident alert system that is used in hilly areas.

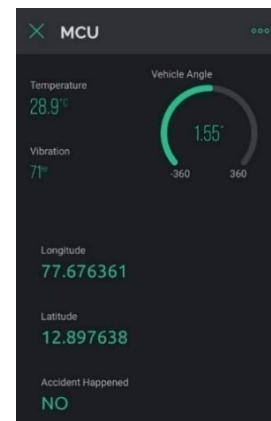


Fig. 5. Thing board App UI (Before Accident)

V. RESULTS

In the data collection phase, the MPU6050 Gyro-Accelerometer becomes instrumental, providing additional layers of information. Figure 6 explains the result obtained in the transmitter side. Both angle and temperature of the car is obtained using MPU6050 sensor. Concurrently, the vibrations of the car are measured using SW-420 Vibration Sensor. Also, Arduino Uno activates the NEO-6M GPS Module to obtain accurate location coordinates, crucial for emergency response. With the collected data in hand, the Arduino Uno communicates with the LoRa Module to transmit the information wirelessly using LoRa technology. The LoRa Module, equipped with the necessary antennas, ensures the efficient transmission of data over extended ranges. On the receiving end, the LoRa Module at the Base Station (Node MCU ESP8266) captures and interprets the transmitted data.

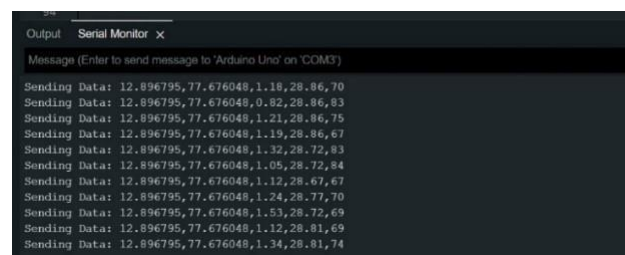


Fig. 6. Arduino IDE Serial Monitor (Transmitter Node)

Upon data reception, the Node MCU ESP8266 determines if the received data indicates a significant event or accident. Figure 7 showcases the receiver node results. The process begins with the continuous monitoring the received values of vibration, angle & temperature of the car.

```
Received packet '12.896795,77.676048,1.34,28.81,74' with RSSI -42
Latitude: 12.896795
Longitude: 77.676048
Angle of vehicle: 1.34° Degrees
Temperature of vehicle: 28.81 °C
Vibration of vehicle: 74 Hz
```

Fig. 7. Arduino IDE Serial Monitor (Receiver Node)

If an accident is confirmed, the system triggers the alert generation process. The GSM Module is activated to send SMS alerts to predefined contacts, promptly notifying relevant individuals or authorities about the accident. Figure 8 showcases the respective result.

```
Alert! Accident Happened...
Sending message to hospital using GSM....
Sending Message
Set SMS Number
SMS Sent!
Sending Message
Set SMS Number
SMS Sent!
Sending Message
Set SMS Number
SMS Sent!
Calling hospital!
```

Fig. 8. SMS alert after accident (Receiver Node)

The Blynk app, serving as the user interface, plays a crucial role in providing real-time visual alerts. As represented in figure 9, the app continuously displaces the obtained values location coordinates, angle, temperature and vibrations of the car on its dashboard. In the case of an accident, the Thingsboard app displays the value of widget “Accident Happened” as “YES”. Emergency responders, alerted through SMS notifications, can then promptly respond to the accident location, aided by the precise GPS coordinates obtained during the incident. Global Positioning System (GPS) technologies determine location by using a network of satellites that orbit the Earth. A GPS receiver calculates its position by precisely timing the signals sent by these satellites. Each satellite continually transmits messages that include the time the message was transmitted, satellite position, and various other data. By calculating the time delay between the transmission and reception of a signal from at least four satellites, the receiver can determine its distance from each satellite and thus its position on Earth (latitude, longitude, and altitude). This method is known as trilateration.

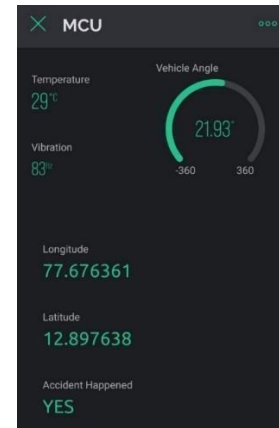


Fig. 9. Thingsboard App UI (After Accident)

Moreover, to enhance the informativeness of the SMS alerts, the system uses obtained accurate location coordinates of the accident site as represented in figure 10. With this information, the system generates a Google map link. The map link, encapsulating the exact accident location, is then included in the SMS alert sent by the GSM Module. Recipients of the SMS alerts, which may include emergency responders and authorities, can interact with the included map link. Clicking on the link within the SMS opens a map interface, pinpointing the precise location of the accident. This visual representation significantly aids in improving the accuracy and efficiency of emergency responses, allowing responders to navigate directly to the incident site.

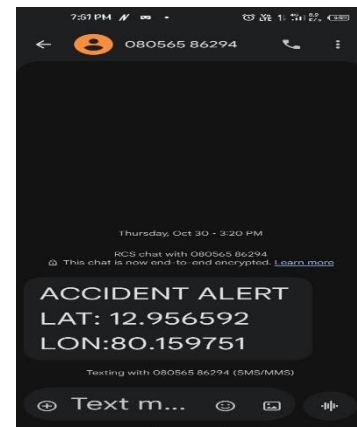


Fig. 10. SMS Alert

The integration of the GSM Module in this system ensures an effective communication by sending SMS alerts with comprehensive details and including a clickable map link, the system provides an enhanced level of information to recipients, facilitating quick and targeted emergency responses in hilly regions.

VI. CONCLUSION

In conclusion, this system's development and evaluation showcases a promising approach for improving safety and reaction mechanisms in difficult terrains. The combination

of LoRa-compatible hardware components such as Arduino, NodeMCU, sensors, and communication modules has resulted in a strong system capable of efficient accident detection and rapid alarm creation. The successful integration of the hardware components enabled reliable accident detection, with the system demonstrating robustness in steep terrains and giving precise location data via the GPS module. LoRa technology was critical to the system's success because it enabled long-distance, low-power communication between the accident node and the base station. This feature was essential for getting beyond geographical barriers, which made the system ideal for installation in hilly regions where traditional communication techniques might not be as effective.

This approach represents a promising path for addressing safety concerns in difficult terrains. Its established performance, strong hardware configuration, and excellent communication mechanisms make it a significant instrument for reducing the impact of accidents in hilly areas. Further developments and adjustments to this technology have the potential to greatly contribute to the field of intelligent transportation systems and emergency response mechanisms. It will include more powerful sensors in terms of their accuracy, latency and durability. Transmission range of LoRa modules can be increased with high range LoRa modules. Also, for achieving more reliability, some methods will be implementing backup communication methods (e.g., GSM, satellite) to ensure alerts are sent even if LoRa fails, incorporating error-checking algorithms in both nodes for verifying data integrity, etc. To improve the existing model, the accuracy of the accident detection can be enhanced by the use of machine learning to filter out the false positives and improve the real time data analysis.

VII. REFERENCES

- [1] Ramesh, P., et al. 2014. "Wireless Sensor Network Development for Landslide Detection." *Ad Hoc Networks* 13: 2–18.
- [2] Deekshit, V.N., et al. 2016. "Smart Geophone Sensor Network for Landslide Signal Detection." In *2016 International Conference on Communication and Signal Processing (ICCCSP)*, 1565–1569.
- [3] Liya, M.L., and M. Aswathy. 2020. "Survey on LoRa Technology for IoT." In *2020 Fourth International Conference on I-SMAC (I-SMAC)*, 8–13.
- [4] Bianco, G.M., et al. 2021. "LoRa System in Mountain Rescue Scenarios." *IEEE Internet of Things Journal* 8(3): 1985–1999.
- [5] Zhang, X., et al. 2021. "Transmission Tower Tilt Monitoring via Low-Power Wide-Area Network." *IEEE Sensors Journal* 21(2): 1100–1107.
- [6] Faber, M.J., et al. 2020. "Evaluation of LoRa Technology Performance." *IEEE Sensors Journal* 20(16): 9480–9489.
- [7] Ramesh, N.V.K., et al. 2020. "Efficient Accident Rescue System using LoRa." *International Journal of Innovative Technology and Exploring Engineering (IJITEE)* 9(6).
- [8] Parveen, N., et al. 2020. "IoT-Based Vehicle Accident Alert System." In *2020 IEEE 5th International Conference on Computing Communication and Automation (ICCCA)*, 330–333.
- [9] Gatade, S., et al. 2022. "Automated Vehicle Accident Detection and IoT Healthcare Alert." In *2022 IEEE 2nd Mysore Sub Section International Conference (MysuruCon)*, 1–5.
- [10] Bhandari, et al. 2023. "IoT-Based Accident & Landslide Detection in Hilly Areas." *Research Reports on Computer Science*.
- [11] Alkhawani, A.H., and B.S. Alsamani. 2023. "Framework for Secure IoT-Based Accident Detection." *Sustainability* 15: 8314.
- [12] R, et al. 2021. "IoT Module for Accident Prevention in Hilly Regions." *International Journal of Engineering Research & Technology (IJERT)* 09(12).
- [13] Nandanija, K., et al. "Automatic Accident Alert System using GSM."
- [14] K.V., et al. 2022. "LoRa-Based Safety Device for Woman Protection." In *2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS)*, 1977–1981.
- [15] G. M, V. C. S, G. Ramasamy and K. B, "Combinatorial Bat Optimization for WSN with Hoover Index Clustering," 2023 *International Conference on Next Generation Electronics (NEleX)*, Vellore, India, 2023, pp. 1-5, doi: 10.1109/NEleX59773.2023.10421682.
- [16] S. K. Sharma, R. K. Saha, and A. Mondal, "Accident detection and vehicle tracking system using IoT-based GSM and GPS technology," *IEEE Sensors Letters*, vol. 6, no. 4, pp. 1–4, 2022.
- [17] M. A. Hossain, S. T. Uddin, and K. Andersson, "Real-time emergency response system using low-power IoT and LPWAN communication," in *Proc. 2021 IEEE Int. Conf. Internet of Things and Intelligence System (IoT&IS)*, pp. 120–125.
- [18] J. R. Karthik and P. P. Kumar, "Performance analysis of LoRa and NB-IoT for remote monitoring in challenging terrains," *IEEE Access*, vol. 9, pp. 145210–145223, 2021.
- [19] L. M. Reza, T. Rahman, and A. O. Lim, "An improved vehicle collision detection algorithm using multi-sensor fusion in IoT," *IEEE Internet of Things Journal*, vol. 8, no. 12, pp. 9875–9886, 2021.
- [20] S. B. Patil, V. N. Kulkarni, and A. G. Ambekar, "LoRa-based low-power wireless monitoring system for emergency alerting in remote areas," in *Proc. 2022 IEEE Int. Conf. Smart Technologies for Smart Nation (SmartTechCon)*, pp. 450–455.